

Abstract No.

A phenotypic drug discovery platform for combinatorial targeting of glioblastoma cell states and entities

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Glioblastoma (GBM) is a highly lethal brain tumor with limited therapeutic options. The blood-brain-barrier (BBB) limits the effectiveness and bioavailability of approved and novel therapeutics. Additionally, GBM consists of heterogeneous, highly plastic cellular entities and states conserved across patients. Tumor cell states adapt in response to standard of care treatment while infiltrating innate immune cells drive a mesenchymal state and acquired drug resistance.

Our lab has developed transcriptional reporters for glioblastoma subtypes named synthetic locus control regions (sLCRs), which inform on cell identity and fate transitions in vitro and in vivo. We combined this tool with four near-isogenic GBM spheroid models representing distinct genotypes found in patients. We established an in vitro phenotypic drug discovery (PDD) platform that recapitulates the GBM – innate immune cell interaction. In this multicellular 3D model, tumor cells and immortalized human microglia co-exist under conditions that enable screening for therapeutic effectiveness of drugs in combination with the standard of care. Our co-culture system uncovers an association between the phenotypic transition and broad resistance to drugs that is distinct to standard neurobasal conditions. We screened 1,241 small molecules, including lipophilic agents which may possess properties to bypass the BBB. We discovered drugs with the distinct ability to modulate tumor cell viability and proliferation, proneural-to-mesenchymal transition and the survival of non-transformed innate immune cells. By using dimensionality reduction, we classified drug responses into four major categories and identified novel responses for a clinically relevant drug with potential therapeutic implications. We assembled data-informed drug combinations by selecting potent candidates from each drug category to target multiple aspects of GBM and identified potent single agents and effective combinations across cell state and identity.

Altogether, our PDD platform identified novel and approved drugs targeting distinct features of the tumor, including a mesenchymal state transition enabling data-driven design of combinatorial treatments.



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